Hypothesis Testing Lecture Exercise

You are a researcher interested in the effect of marine protected areas on fish populations. To begin your investigation, you decide to take a look at weights of Yellowfin Tuna living within and outside of a marine protected area (MPA). You randomly sample 40 tuna inside and 40 tuna outside of an MPA, and take their weights (kg).

Make a copy of this spreadsheet (https://docs.google.com/spreadsheets/d/14Aowo3HJJGGQgWPepTyO0dkTPEosliyygdOnC9PEzkk/edit?usp=sharing) to your own Google Drive, and use it to complete this activity.

Tasks:

- Fill out the empty boxes in the spreadsheet.
    - For the first three sets, you will want to use the built in formulas Sheets has for the average, standard deviation, and variance.
    - The fourth set is filled out for you.
    - For the three boxes after that, replace the Xs in the typed formulas with the appropriate cells (i.e., A1). You can delete the X and click on the cell you want to use in that spot. It's recommended to type in the editing bar underneath the undo/redo/print buttons, and to move your cursor to a different spot in the formula by clicking once you have entered the cell number you would like.
    - For the last box, use this table (https://www.itl.nist.gov/div898/handbook/eda/section3/eda3672.htm), but be sure to read the numbered instructions on the webpage!
- Briefly answer the following questions:
    - State an appropriate null and alternative hypothesis.
    - We randomly sampled our tuna, and we are going to assume a normal distribution for the sake of the person writing this assignment. Have we met the third assumption of our standard t-test? How do you know?
    - Is the test statistic that we calculated greater than or less than the corresponding critical value? Given this, should we reject our null hypothesis?

Deliverables:

- A screenshot of your results on your spreadsheet (5 points)
- A sharable link to your spreadsheet
- Answers to the questions listed above (3 points)

You can put all of these things onto one PDF/word document and submit it if that's the easiest thing for you to do!

This is due on Sunday, 09/13 at 11:59 PM.